

OBJECT ORIENTED PROGRAMMING

#2

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EXERCISE #1

```
#include <iostream>
```

```
using namespace std;
```

```
void sort3(double& a, double& b, double& c, bool (*compare)(double&, double&));
```

```
void print3(const double& a, const double& b, const double& c);
```

```
void swap2(double& a, double& b);
```

```
bool ascending(double& a, double& b);
```

```
bool descending(double& a, double& b);
```

```
void sort3(double& a, double& b, double& c, bool (*compare)(double&, double&)) {
```

```
    if (!compare(a, b)) {
```

```
        swap(a, b);
```

```
    }
```

```
    if (!compare(b, c)) {
```

```
        swap(b, c);
```

```
    }
```

```
    if (!compare(a, b)) {
```

```
        swap(a, b);
```

```
    }
```

```
}
```

```
void print3(const double& a, const double& b, const double& c) {  
    cout << a << " " << b << " " << c << endl;  
}
```

```
void swap2(double& a, double& b) {  
    double temp = a;  
    a = b;  
    b = temp;  
}
```

```
bool ascending(double& a, double& b) {  
    return a < b;  
}
```

```
bool descending(double& a, double& b) {  
    return a > b;  
}
```

```
int main() {  
    double na1 = 10.3, na2 = -2.1, na3 = 8.0;  
    double nd1 = 10.3, nd2 = -2.1, nd3 = 8.0;  
  
    cout << "Before sort: ";  
    print3(na1, na2, na3);  
  
    sort3(na1, na2, na3, ascending);
```

```
sort3(nd1, nd2, nd3, descending);

cout << "After sort(ascending): ";

print3(na1, na2, na3);

cout << "After sort(descending): ";

print3(nd1, nd2, nd3);

return 0;

}
```

```
Before sort: 10.3 -2.1 8
After sort(ascending): -2.1 8 10.3
After sort(descending): 10.3 8 -2.1
```